Journal Recommendation System and Topic Clustering for Computer Science Articles

Data Mining Final Project

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Final delivery regenerated from the accepted $K = 46$ pipeline

***Abstract*—This project implements a journal recommendation and topic clustering pipeline on the *CompSciencePub.sqlite* database. The recommendation stage joins abstracts with author keywords, Web of Science subjects, and keyword plus fields, then ranks journals with a TF-IDF and cosine-similarity baseline. A leakage-free stratified 80/20 hold-out split over journals with at least five articles produced 28.37% Top-1 accuracy, 45.89% Top-3 accuracy, 54.96% Top-5 accuracy, and 0.410 mean reciprocal rank on 4,596 held-out articles. For topic discovery, a domain-aware TF-IDF space was reduced with Truncated SVD and normalized before K-Means. Instead of choosing K from a small local sweep, the final delivery analyzes $K = 2..100$ with step 2 plus the manually compared $K = 45$; in the current reproducible run, the Kneedle algorithm identifies $K = 46$ as the elbow point. Although larger K values achieve better scores on some individual metrics, the final submission keeps the $K = 46$ K-Means model because it provides a balanced trade-off between clustering quality, model complexity, cluster-size distribution, and topic interpretability. An experimental BERTopic alternative was also evaluated with TF-IDF/SVD and sentence-transformer embeddings, but it was not selected for the final delivery because it depends on a heavier experimental stack and assigns a noticeable fraction of documents to outliers.**

***Index Terms*—Data Mining, Text Mining, Recommender Systems, TF-IDF, K-Means, BERTopic, Topic Clustering**

I. Introduction

Selecting a suitable journal for a new manuscript is difficult because many journals overlap in topic, methodology, and audience. A practical journal finder should therefore rely on the manuscript content itself. In parallel, topic clustering can reveal the broad structure of the same literature collection and help explain how the corpus is organized.

The homework requires two core capabilities: recommending the top five journals for a user-provided abstract, and producing meaningful topic clusters from the provided computer science publication database. The final delivery also documents how the clustering decision was made, which alternatives were explored, and why the final method was preferred.

II. Literature Review

Academic journal recommendation is commonly modeled as a content-based retrieval problem. TF-IDF remains a strong baseline for scientific text, especially when abstracts are enriched with controlled metadata such as author keywords and subject terms [1], [2]. The ranking challenge is not only article retrieval but also journal-level aggregation of article similarities [3].

For unsupervised topic discovery, K-Means is still attractive because it is fast, reproducible, and straightforward to interpret through cluster centroid terms. Truncated SVD is frequently paired with sparse text features to compress the term space while preserving major semantic structure [4]. More recent topic-modeling pipelines such as BERTopic combine document embeddings, manifold learning, density-based clustering, and c-TF-IDF topic representations, but they often trade simplicity and determinism for additional flexibility.

III. Methodology

A. Data Integration and Preprocessing

The database tables *AcademicRecord*, *AcademicRecordAbstract*, and *Publication* were joined to obtain article identifiers, abstracts, and journal names. Additional one-to-many metadata from author keywords, Web of Science subjects, and keyword plus terms were concatenated into a single textual field per article. The text was then lowercased, stripped of HTML tags, purged of non-alphabetic characters, and normalized for whitespace.

B. Journal Recommendation Model

The recommender uses *TfidfVectorizer* with English stop-word removal, a maximum of 10,000 features, minimum document frequency of 2, and unigram-bigram features. Given a query abstract, cosine similarity is computed against all indexed articles. Journal scores are aggregated as

$$Score(J) = 0.7 \times \max(sim_J) + 0.3 \times \mathrm{mean}(sim_J). \quad (1)$$

The system returns the five journals with the highest scores.

C. Evaluation Protocol

Recommendation quality is measured with a stratified 80/20 hold-out split over journals that contain at least five articles. TF-IDF is fit only on the training partition, and the held-out articles are queried against that index. Performance is reported using Top-1, Top-3, Top-5 accuracy, and mean reciprocal rank (MRR).

D. Clustering Representation

The clustering feature space uses a second TF-IDF vectorizer dedicated to topic discovery. Besides standard English stop words, it removes domain-generic research terms such as *paper*, *method*, *system*, *computer*, and *science*. The vectorizer uses 15,000 maximum features, minimum document frequency of 5, maximum document frequency of 0.35, and unigram-bigram features. The sparse TF-IDF matrix is then reduced to 50 latent dimensions with Truncated SVD and L_2 normalized before K-Means.

E. K Selection Analysis

The final delivery no longer chooses K from a small local sweep. Instead, the clustering feature space is evaluated over $K = 2..100$ with step 2, plus the manually compared $K = 45$. For each candidate K, inertia, silhouette score, Davies-Bouldin score, Calinski-Harabasz score, and cluster-size statistics are recorded. In the current reproducible run, the Kneedle algorithm identified $K = 46$ as the elbow point from the inertia curve. Although $K = 54$ achieved the best silhouette score and $K = 62$ achieved the best Davies-Bouldin score, $K = 46$ was kept as the final operating point because it provides a balanced trade-off between clustering quality, model complexity, cluster-size distribution, and topic interpretability.

F. Alternative Method Explored

As an experimental alternative, BERTopic was tested with two document-embedding backends: a TF-IDF plus TruncatedSVD embedding and a sentence-transformer embedding based on the all-MiniLM-L6-v2 model. The BERTopic pipeline used UMAP for dimensionality reduction, HDBSCAN for density-based clustering, and c-TF-IDF for topic representation. This alternative was kept outside the main delivery path and used only as a comparison method.

IV. Results and Discussion

A. Dataset Summary

Table I summarizes the corpus used by the final pipeline.

TABLE I
DATASET SUMMARY

Statistic	Value
Articles with abstracts	23,061
Distinct journals	455
Mean abstract length	164.85 words
Median abstract length	159 words
Missing author keywords	14.81%
Missing subjects	0.00%
Missing keyword plus	23.65%

B. Recommendation Performance

Table II reports the leakage-free hold-out performance. The Top-5 accuracy of 54.96% means that the correct journal appears in the recommendation list for more than half of the held-out articles.

TABLE II
HOLD-OUT RECOMMENDATION METRICS

Split	Top-1	Top-3	Top-5	MRR
80/20 hold-out	0.2837	0.4589	0.5496	0.4102

TABLE III
TOP-5 JOURNAL RECOMMENDATIONS FOR THE SAMPLE ABSTRACT

Rank	Journal	Score
1	ACM SIGCOMM Computer Communication Review	0.2432
2	Knowledge and Information Systems	0.2353
3	Frontiers of IT & Electronic Engineering	0.2343
4	IEEE Wireless Communications	0.2336
5	Int. J. Computers Communications & Control	0.2327

TABLE IV
FINAL K COMPARISON AROUND THE ACCEPTED CHOICE

K	Silhouette	Inertia	DB	Min size
44	0.1859	9214.36	1.7508	116
45	0.1816	9174.05	1.7661	150
46	0.1848	9058.07	1.7484	118
48	0.1864	8930.96	1.7651	152
50	0.1896	8747.15	1.6887	117

C. Sample Journal Recommendation

To make the final journal finder behavior explicit, the sample abstract below was queried against the trained recommender:

> This paper proposes a machine learning based method for detecting anomalies in network traffic using deep neural networks and feature selection techniques.

Table III lists the top five journals returned for this abstract.

D. K Selection and Final Choice

Table IV shows the strongest comparison points around the final decision. In the current reproducible run, the Kneedle algorithm identified $K = 46$ as the elbow point on the inertia curve. This does not mean that $K = 46$ is the best value for every metric: $K = 54$ achieved the best silhouette score and $K = 62$ achieved the best Davies-Bouldin score. However, $K = 46$ was retained as the final operating point because it keeps cluster quality strong while avoiding unnecessary model fragmentation.

E. Alternative Method Results

Table V summarizes the BERTopic alternatives. Both variants produced reasonable topic groups, but they also assigned

TABLE VI
REPRESENTATIVE CLUSTERS IN THE FINAL $K = 46$ MODEL

ID	Interpreted topic from top terms	Size
1	Routing and ad hoc networking: routing, ad hoc, protocols, telecommunications	391
8	Network traffic and optical networks: network, traffic, optical, bandwidth	634
13	Programming languages and verification: language, programming, semantics, verification	552
15	Wireless communication and MIMO systems: channel, spectrum, fading, MIMO	696
20	Image analysis: image, segmentation, color, imaging, texture	766
23	Mobile and recommender systems: user, mobile, context, recommender, recommendation	561
26	Cloud computing and services: cloud computing, service, resource, security	466
32	Software engineering and requirements: software engineering, quality, requirements, code	737
36	Databases, data mining, and retrieval: mining, query, retrieval, databases	729
41	Security and authentication: security, authentication, attack, protocol, secure	651
42	Software testing and fault coverage: testing, fault, test case, coverage	264
43	Graph algorithms and approximation: graph, vertex, approximation, algorithms	535

a noticeable fraction of documents to outliers. The sentence-transformer version produced fewer topics but a larger outlier ratio. BERTopic was not used in the final submission because it depends on a heavier experimental toolchain, introduces density-based outlier handling that leaves many documents unassigned, and is less straightforward to explain than the final deterministic $K = 46$ K-Means pipeline.

TABLE V
BERTOPIC ALTERNATIVE SUMMARY

Method	Topics	Outliers	Outlier ratio
BERTopic (TF-IDF + TruncatedSVD)	52	3234	14.02%
BERTopic (sentence-transformer)	41	4449	19.29%

F. Representative Clusters from the Final $K = 46$ Model

The selected $K = 46$ model produces interpretable clusters that map well to recognizable computer science subfields. Table VI lists representative examples from networking, communications, software engineering, data management, and security.

V. CONCLUSION

The final delivery satisfies the homework objectives while presenting a cleaner and more defensible clustering decision process. The recommendation stage remains a strong classical baseline with solid hold-out performance. The clustering stage now documents a broad K search, retains the accepted elbow-based $K = 46$ decision, and clearly distinguishes the final K-Means solution from the experimental BERTopic alternatives.

The final choice favors reproducibility, interpretability, and a lightweight submission over a heavier experimental topic-modeling stack. This makes the delivered project easier to run, easier to explain, and better aligned with the homework requirement of producing a practical journal finder and topic clustering system.

REFERENCES

[1] P. Lops, M. de Gemmis, and G. Semeraro, "Content-based Recommender Systems: State of the Art and Trends," in *Recommender Systems Handbook*, Springer, 2011, pp. 73–105.

[2] X. Wang et al., "A literature review on academic recommender systems," *Information Processing & Management*, vol. 54, no. 6, pp. 883–912, 2018.

[3] S. Sugiyama and M. Kan, "Scholarly Paper Recommendation via User's Recent Research Interests," in *Proc. of the 10th ACM/IEEE-CS Joint Conference on Digital Libraries*, 2010.

[4] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*, Cambridge University Press, 2008.

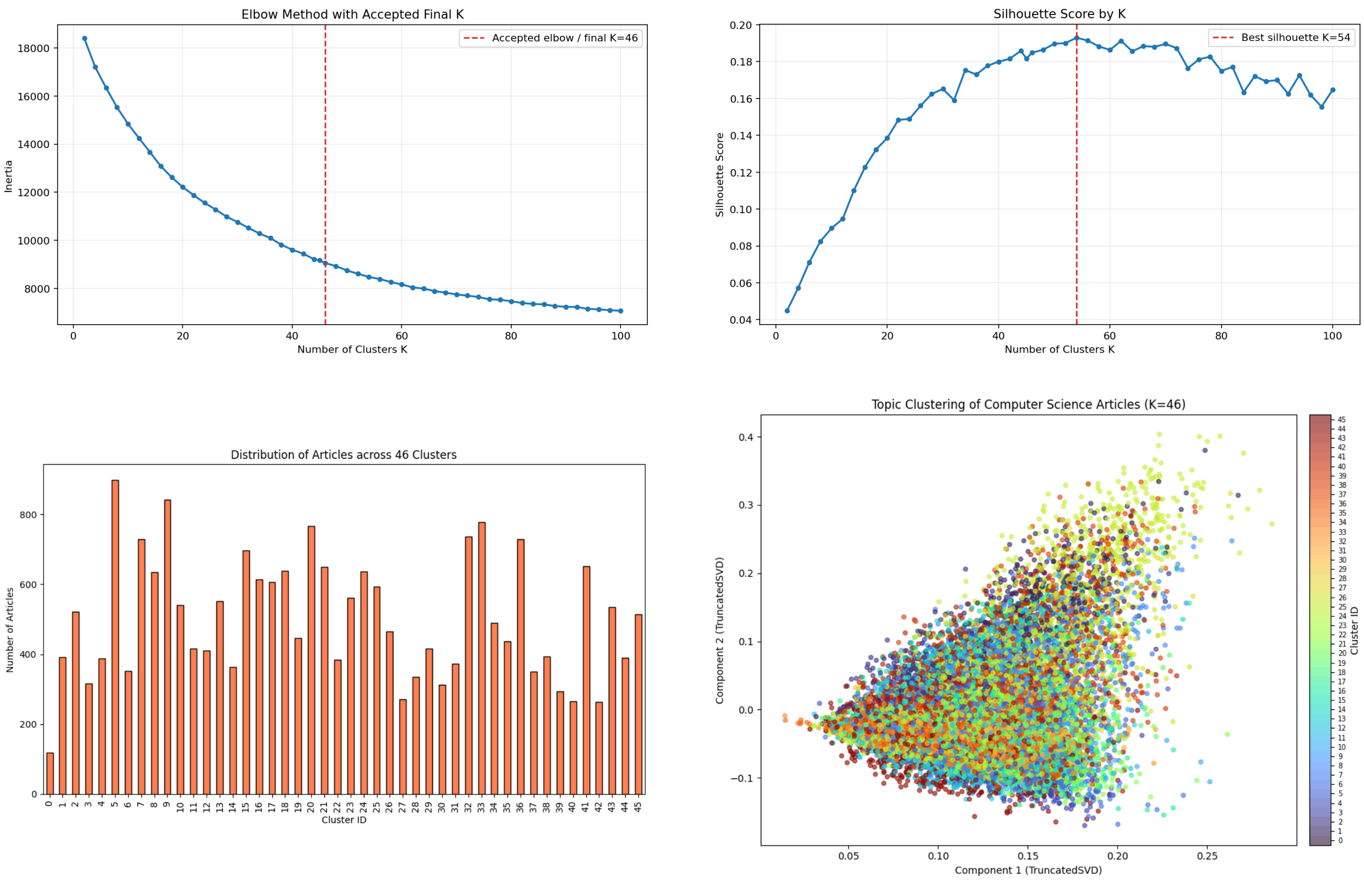

Fig. 1. Final visual outputs. Top-left: elbow / inertia curve with the accepted final choice. Top-right: silhouette score across the broad K sweep. Bottom-left: article counts per cluster for the accepted $K = 46$ model. Bottom-right: two-dimensional Truncated SVD projection of the final clustered feature space.